Industrial Internship Report on

"Automotive E-Commerce System"

Prepared by

Soumyaditya Majhi

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Automotive E-Commerce System", where I worked on designing and developing a system for managing products, understanding frontend and backend integration, and implementing database connectivity using Java and JDBC. The project included concepts of full stack development, object-oriented programming, and basic web technologies. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface 3

2 Introduction 5

2.1 About UniConverge Technologies Pvt Ltd 5

2.2 About upskill Campus 9

2.3 Objective 11

2.4 Reference 11

2.5 Glossary 11

3 Problem Statement 12

4 Existing and Proposed solution 13

5 Proposed Design/ Model ..... **Error! Bookmark not defined.**

5.1 High Level Diagram (if applicable) ..... **Error! Bookmark not defined.**

5.2 Low Level Diagram (if applicable) ..... **Error! Bookmark not defined.**

5.3 Interfaces (if applicable) ..... **Error! Bookmark not defined.**

6 Performance Test ..... **Error! Bookmark not defined.**

6.1 Test Plan/ Test Cases ..... **Error! Bookmark not defined.**

6.2 Test Procedure ..... **Error! Bookmark not defined.**

6.3 Performance Outcome ..... **Error! Bookmark not defined.**

7 My learnings ..... **Error! Bookmark not defined.**

8 Future work scope ..... **Error! Bookmark not defined.**

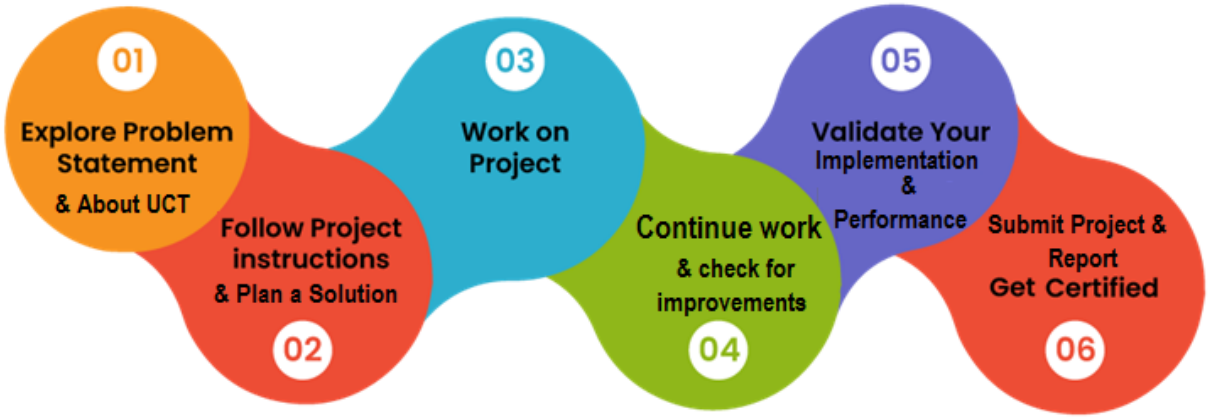
1 Preface

This report summarizes the work completed during the 6 weeks of Industrial Internship under upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). Throughout the internship, I worked on understanding and implementing concepts related to Full Stack Web Development and Java programming.

Internships play a very important role in career development as they provide practical exposure to real-world problems and help bridge the gap between theoretical knowledge and industry requirements. This internship helped me gain hands-on experience and improved my technical and problem-solving skills.

The project assigned to me was "Automotive E-Commerce System", which focuses on managing products and understanding the integration of frontend, backend, and database systems using Java and basic web technologies.

This internship provided a great opportunity to learn under the guidance of USC/UCT and understand how real industry projects are planned and executed. The program was well structured, starting from basic concepts and gradually moving towards advanced topics like OOP, JDBC, and Full Stack Development.Overall, the internship was well planned with weekly tasks, continuous learning, and practical implementation, which helped me build a strong foundation in development and improved my confidence in handling real-world projects.



Your Learnings and overall experience.

During this internship, I gained a strong understanding of Full Stack Web Development along with Java programming concepts. I learned core Java, object-oriented programming principles, advanced Java

features such as streams and lambda expressions, and database connectivity using JDBC. I also explored frontend basics like HTML and understood how frontend and backend components work together in a real-world application. Overall, this internship improved my coding skills, problem-solving ability, and confidence in building projects.

Thank to all, who have helped you directly or indirectly.

I would like to express my sincere gratitude to upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) for providing me this valuable opportunity. I am thankful to all the mentors, instructors, and coordinators who guided me throughout the internship. Their support and guidance helped me overcome challenges and improve my understanding. I would also like to thank my friends and peers who supported me during this journey.

I would like to advise my juniors and peers to actively participate in internships and focus on practical learning. Consistent practice, curiosity to learn new technologies, and patience while solving problems are key to success. Make the most out of such opportunities as they help in building strong fundamentals and preparing for real-world challenges.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

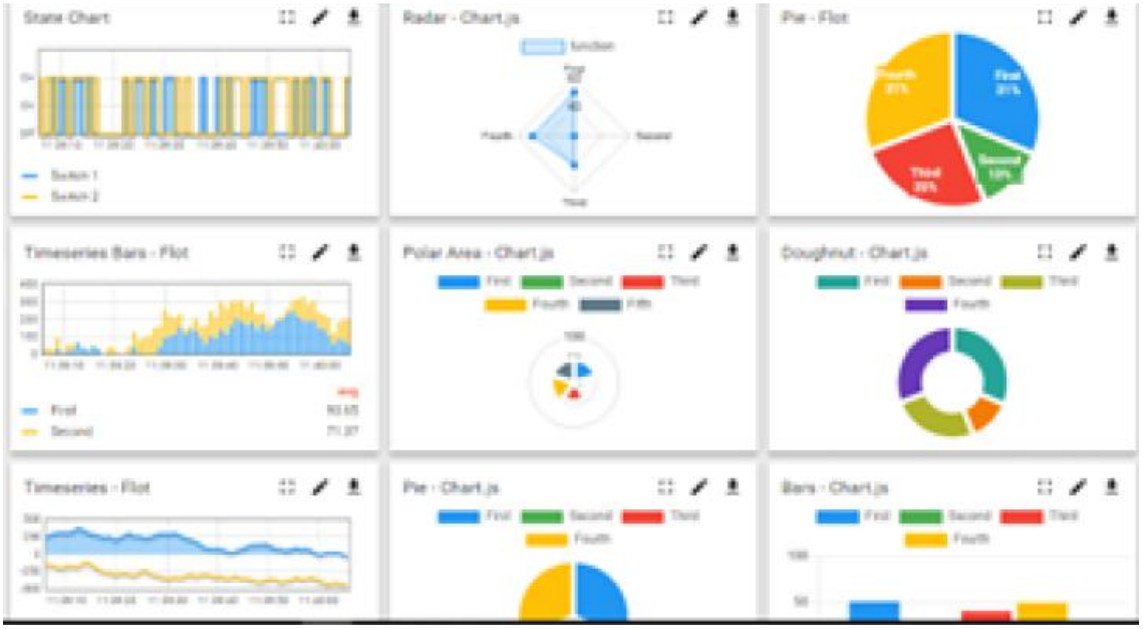
i. UCT IoT Platform (UCT Insight)

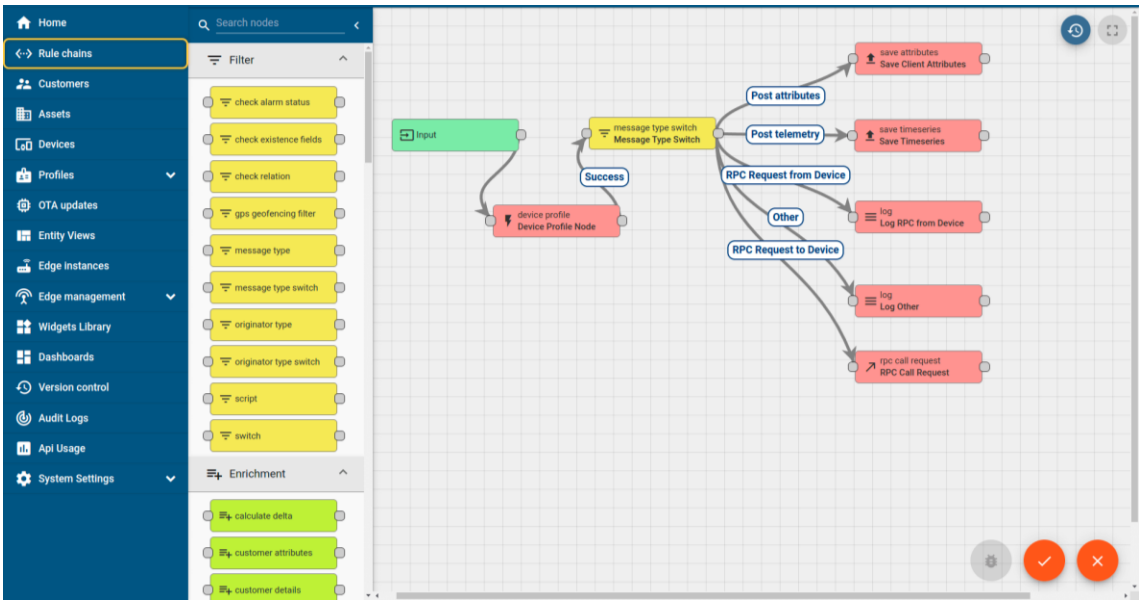
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

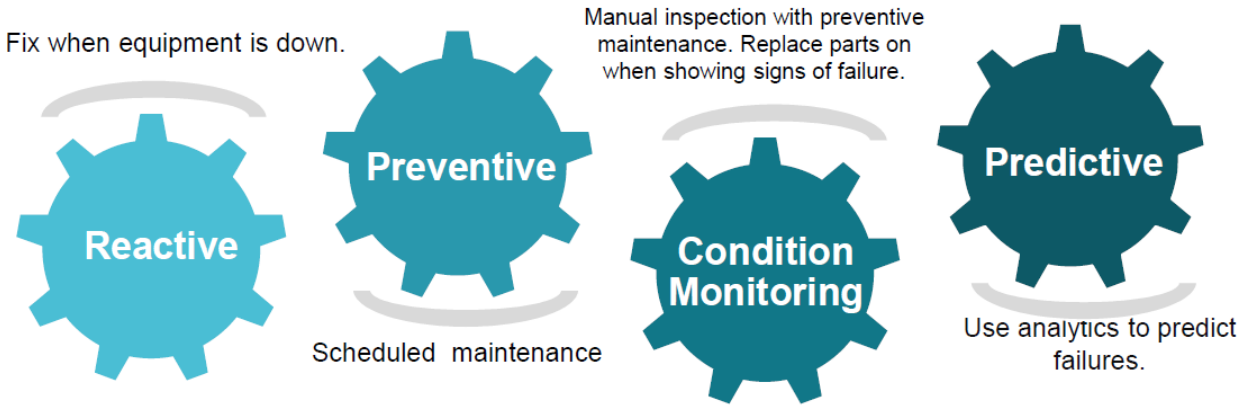
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

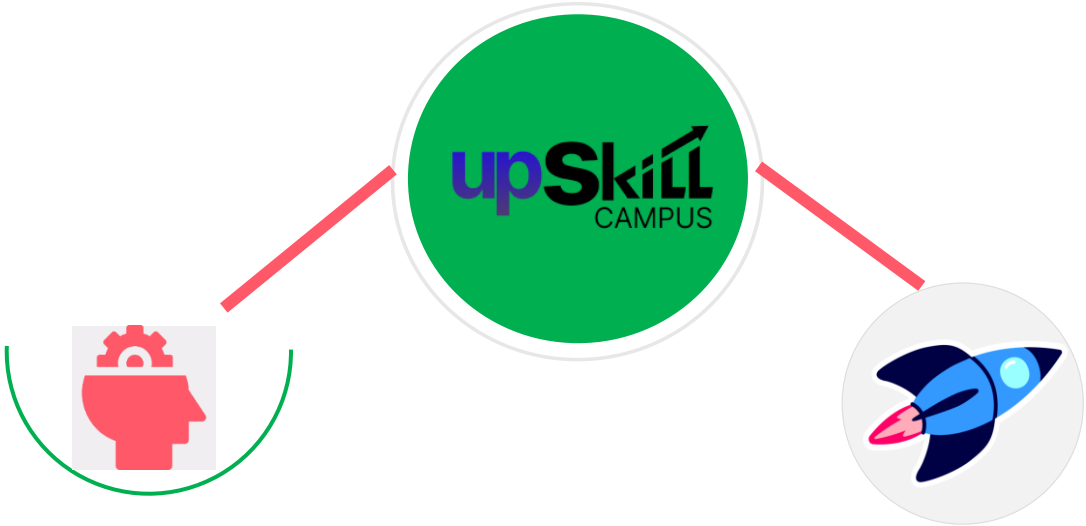
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

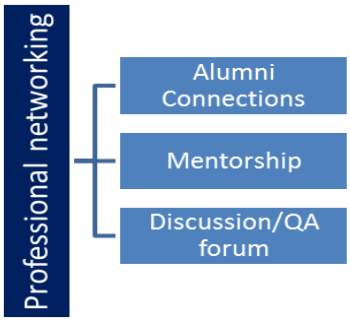
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

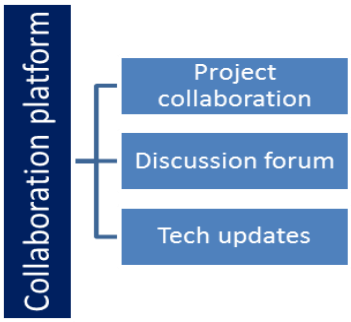


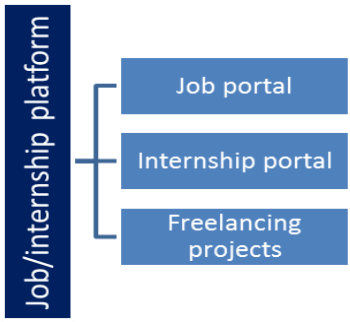
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Full Stack Web Development Course Material provided by upskill Campus and The IoT Academy

[2] Official Java Documentation – https://docs.oracle.com/javase/tutorial/

[3] W3Schools – https://www.w3schools.com/

2.6 Glossary

Terms	Acronym
FullStack Development	FSD
Java Database Connectivity	JDBC
Object-Oriented Programming	OOP

3 Problem Statement

In the assigned problem statement, the objective was to design and develop an Automotive E-Commerce System that simulates real-world online shopping functionality. The system focuses on managing product information, enabling users to view and interact with available products, and performing basic operations such as adding and retrieving product details from a database.

The project also aims to demonstrate the integration of different technologies learned during the internship, including Java for backend logic, Object-Oriented Programming concepts for structured design, and JDBC for database connectivity. Additionally, the project reflects an understanding of frontend basics and how they connect with backend systems in a full stack application.

The problem statement emphasizes building a scalable and structured solution that can be extended in the future with features like user authentication, cart management, and payment integration.

4 Existing and Proposed solution

Existing solutions for e-commerce systems mainly include large-scale platforms that provide product listing, user interaction, and transaction management. These systems are highly efficient but often complex and resource-intensive. For beginners, such systems can be difficult to understand and implement due to their large codebase, advanced frameworks, and multiple integrated services. Additionally, many basic implementations available online focus only on frontend design and lack proper backend integration or database handling.

The proposed solution in this project is to develop a simplified Automotive E-Commerce System using Java and JDBC. The system is designed to handle product management operations such as adding and viewing products while maintaining a clear and modular structure using Object-Oriented Programming principles. It focuses on building a strong foundation by integrating backend logic with database connectivity.

The value addition of this project lies in its simplicity and clarity, making it suitable for learning and demonstration purposes. It provides a practical understanding of how real-world applications manage data and interact with databases. The project is also scalable and can be extended in the future by adding features such as user authentication, shopping cart functionality, payment systems, and frontend integration, thereby evolving into a complete full stack application.

4.1 Code submission (Github link)

https://github.com/Soumyditya/upskillcampus.git

4.2 Report submission (Github link) :

https://github.com/Soumyditya/upskillcampus.git

5 Proposed Design/ Model

The proposed design of the Automotive E-Commerce System follows a structured and modular approach, focusing on clear separation between different components of the application. The system is designed using Object-Oriented Programming principles and integrates database operations using JDBC.

The design flow of the system can be understood in three main stages:

Start Stage:

In the initial stage, the system begins with user interaction through a simple interface (console-based). The user is provided with options such as adding new products and viewing existing products. This stage focuses on taking input and initiating operations.

Intermediate Stage:

In this stage, the application processes the user input using backend logic written in Java. The Product class is used to represent product data, and the ProductDAO class handles all database-related operations. JDBC is used to establish a connection between the application and the database. The system performs operations such as inserting product details into the database and retrieving stored data.

Final Outcome Stage:

In the final stage, the processed data is displayed back to the user. The system successfully shows product details retrieved from the database, ensuring proper data handling and output. This demonstrates the working of a basic e-commerce system where backend processing and database interaction are effectively implemented.

Overall, the design ensures scalability and flexibility, allowing future enhancements such as frontend integration, user authentication, shopping cart functionality, and payment systems to transform it into a complete full stack application.

6 Performance Test

This section highlights how the developed system aligns with real-world industrial requirements rather than being just an academic implementation. The Automotive E-Commerce System was evaluated based on certain practical constraints such as performance, scalability, and data handling efficiency.

One of the primary constraints considered was database performance and response time. Since the system uses JDBC for database connectivity, efficient query execution was important. This was handled by using PreparedStatement, which improves performance and prevents SQL injection.

Another constraint was memory usage and system efficiency. The application is designed using simple and modular classes, ensuring that only required data is processed and stored. This reduces unnecessary memory consumption and keeps the system lightweight.

Speed of operations was also considered. Basic operations such as inserting and retrieving product data were tested, and the system was able to perform these tasks efficiently with minimal delay under normal conditions.

In terms of scalability, the current system is a basic implementation, but the design allows future enhancements. The modular structure ensures that additional features like user authentication, cart system, and payment integration can be added without affecting the existing functionality.

Although advanced performance metrics like MIPS, power consumption, and durability were not practically measured, their impact was analyzed. For large-scale systems, database optimization, indexing, and efficient query handling would be required. It is recommended to use connection pooling, caching mechanisms, and optimized database design for better performance in real-world scenarios.

Overall, the system demonstrates a foundational approach to handling real-world constraints and provides a scalable base for further development into a complete industrial-level application.

6.1 Test Plan / Test Cases

The test plan for the Automotive E-Commerce System focuses on verifying the correctness and efficiency of core functionalities such as product insertion, data retrieval, and database connectivity.

Test Case 1:

Test Objective: Verify product insertion into the database

Input: Product ID, Name, and Price entered by user

Expected Output: Product should be successfully added to the database

Result: Product added successfully and confirmation message displayed

Test Case 2:

Test Objective: Verify retrieval of product data

Input: User selects "View Products" option

Expected Output: List of all products stored in the database should be displayed

Result: All stored product details displayed correctly

Test Case 3:

Test Objective: Validate database connectivity

Input: Establish connection using JDBC

Expected Output: Successful connection to the database without errors

Result: Connection established successfully

Test Case 4:

Test Objective: Handle invalid input

Input: Incorrect or empty product details

Expected Output: System should handle error gracefully without crashing

Result: Error handled and appropriate message displayed

Test Case 5:

Test Objective: Check system performance for multiple entries

Input: Insert multiple product records sequentially

Expected Output: System should handle multiple insert operations efficiently

Result: System performed operations without significant delay

Overall, the test cases confirm that the system performs its intended operations correctly and handles basic scenarios effectively.

6.2 Test Procedure

The testing of the Automotive E-Commerce System was carried out in a systematic manner to ensure that all functionalities work as expected. The following steps were followed during the testing process:

Step 1:

The application was compiled and executed using a Java development environment. All required files and dependencies were properly configured before running the program.

Step 2:

Database connectivity was tested by establishing a connection using JDBC. The connection parameters such as database URL, username, and password were verified to ensure successful communication with the database.

Step 3:

The product insertion functionality was tested by entering valid product details such as ID, name, and price. The data was then checked in the database to confirm successful storage.

Step 4:

The product retrieval functionality was tested by selecting the option to view products. The system fetched and displayed all stored records, which were verified against the database entries.

Step 5:

Error handling was tested by providing invalid or incomplete inputs. The system behavior was observed to ensure that it handled such cases without crashing.

Step 6:

Multiple test cases were executed repeatedly to check consistency and stability of the system. The application maintained correct functionality throughout the testing process.

The above procedure ensured that all major components of the system were tested and validated properly.

6.3 Performance Outcome

The performance outcome of the Automotive E-Commerce System indicates that the application functions efficiently under basic operational conditions. The system was able to successfully perform key operations such as inserting and retrieving product data from the database with minimal delay.

The response time for database operations was observed to be fast for small to moderate data inputs. The use of JDBC and PreparedStatement ensured efficient execution of SQL queries and improved overall performance.

The system demonstrated stability during repeated test executions, handling multiple operations without any major errors or crashes. Memory usage remained minimal due to the lightweight design and efficient handling of data structures.

Although the system is currently designed for small-scale use, it provides a solid foundation for further scalability. With proper enhancements such as database indexing, connection pooling, and optimized query handling, the system can be extended to handle larger datasets and real-world workloads.

Overall, the performance of the system meets the expected requirements for a basic e-commerce application and validates its functionality as a reliable and scalable solution for future development.

7 My learnings

During this internship, I gained comprehensive knowledge of Full Stack Web Development and Java programming. I started with understanding project requirements and gradually learned core Java concepts, Object-Oriented Programming principles, and advanced Java features such as streams and lambda expressions. I also learned how to connect Java applications with databases using JDBC and perform operations like data insertion and retrieval.

In addition to backend development, I explored basic frontend technologies like HTML, which helped me understand how user interfaces interact with backend systems in a full stack application. This internship also improved my problem-solving skills, debugging ability, and logical thinking through continuous practice and implementation.

Overall, the learning experience has significantly contributed to my technical growth and prepared me for real-world software development. The skills acquired during this internship will help me in building scalable applications and will be highly beneficial for my future career in the field of software development.

8 Future work scope

Due to time limitations, the current project focuses on basic functionalities of an Automotive E-Commerce System. However, there are several enhancements that can be implemented in the future to make the system more advanced and closer to real-world applications.

One of the major improvements would be the integration of a proper frontend interface using technologies like HTML, CSS, and JavaScript to make the system user-friendly and interactive. Additionally, implementing user authentication features such as login and registration would enhance security and usability.

The system can also be extended by adding a shopping cart feature, order management system, and payment gateway integration to simulate a complete e-commerce platform. Another important enhancement would be optimizing the database by applying indexing and improving query performance for handling large-scale data.

Furthermore, the project can be upgraded to a full stack web application using frameworks and deployed on cloud platforms for real-time access. Advanced features such as product search, filtering, and recommendation systems can also be incorporated.

These improvements will help in transforming the current basic system into a fully functional, scalable, and industry-level application in the future.